

- Business Requirements Document (BRD)**
- Project Title:** Data Integration and Reporting for Customer and Order Data
- Document Version:** 1.0
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- Date:** [Current Date]
- Introduction:**

The purpose of this document is to outline the business requirements for a data integration and reporting project involving customer and order data. The project aims to integrate data from CSV files into a Delta table, perform data cleansing and transformation, and create aggregated reports.

- Business Requirements:**

1. **Data Ingestion**

- Read source CSV data from the specified volumes:
`/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/customerdata` and
`/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/orderdata`.
- Load the data into Delta tables `customer` and `order`.

2. **Data Schema**

- The schema for the `customer` table is: `CustId`, `Name`, `EmailId`, `Region`.
- The schema for the `order` table is: `OrderId`, `ItemName`, `PricePerUnit`, `Qty`, `Date`, `CustId`.

3. **Data Cleansing**

- Remove "Null"/Null and duplicate records from both `customer` and `order` tables.

4. **Data Transformation and Loading**

- Create an `ordersummary` table if it does not exist in the catalog "gen_ai_poc_databrickscoe" and schema "sdlc_wizard" with the schema: `CustId`, `Name`, `EmailId`, `Region`, `OrderId`, `ItemName`, `PricePerUnit`, `Qty`, `Date`.
- Join the `customer` and `order` data using the "CustId" field and load the data into an SCD Type 2 table `ordersummary`.
- Include logic to update the SCD Type 2 table `ordersummary` whenever there is a change in the `customer` table. Old records should be made inactive, and new records should be made active. Update `StartDate` and `EndDate` accordingly.

5. **Aggregated Reporting**

- Create a new table `customeraggregatespend` with columns "Name", "TotalAmount", and "Date" in the catalog "gen_ai_poc_databrickscoe" and schema "sdlc_wizard" if it does not exist.

- Aggregate the "TotalAmount" column from the `ordersummary` table and group it by the "Name" and "Date" columns. Load the aggregated data into the `customeraggregatespend` table.

- Functional Requirements:**

1. The system shall read CSV data from the specified volumes and load it into Delta tables.
2. The system shall perform data cleansing by removing "Null"/Null and duplicate records.
3. The system shall create the `ordersummary` table if it does not exist and load the joined data into an SCD Type 2 table.
4. The system shall update the SCD Type 2 table `ordersummary` whenever there is a change in the `customer` table.
5. The system shall create the `customeraggregatespend` table if it does not exist and load aggregated data into it.

- Non-Functional Requirements:**

1. **Performance:** The system shall process the data within a reasonable time frame to meet business needs.
2. **Data Quality:** The system shall ensure data accuracy, completeness, and consistency.
3. **Security:** The system shall adhere to data security and access controls as per organizational policies.

- Assumptions and Dependencies:**

1. The CSV files are in the correct format and are available at the specified locations.
2. The necessary catalog and schema exist in the Databricks environment.
3. The required permissions and access controls are in place.

- Acceptance Criteria:**

1. The data is successfully ingested into Delta tables.
2. The data is cleansed and transformed as per the requirements.
3. The `ordersummary` table is created and updated correctly.
4. The `customeraggregatespend` table is created and populated with aggregated data.

- Risks and Mitigation:**

1. **Data Quality Issues:** Implement data validation and cleansing checks to mitigate data quality issues.

2. ****System Performance:**** Monitor system performance and optimize as needed to ensure timely processing.

By following this BRD, the project stakeholders can ensure that the data integration and reporting project meets the business requirements and is delivered within the expected timelines and budget.